

635 Course Review

Time Harmonic Maxwell's Equations

$$\nabla \times \mathbf{E} = -j\omega\mu\mathbf{H} - \mathbf{M}$$

$$\nabla \times \mathbf{H} = j\omega\epsilon\mathbf{E} + \mathbf{J}$$

$$\nabla \cdot \mathbf{E} = \rho / \epsilon$$

$$\nabla \cdot \mathbf{H} = \rho_m / \mu$$

constitutive relations:

$$\mathbf{D} = \epsilon\mathbf{E}$$

$$\mathbf{B} = \mu\mathbf{H}$$

Helmholtz Equation

$$\nabla^2 \mathbf{E} + k^2 \mathbf{E} = j\omega\mu\mathbf{J}^i - \frac{\nabla(\nabla \cdot \mathbf{J}^i)}{\sigma + j\omega\epsilon} + \nabla \times \mathbf{M}^i$$

Free Space Helmholtz Equation

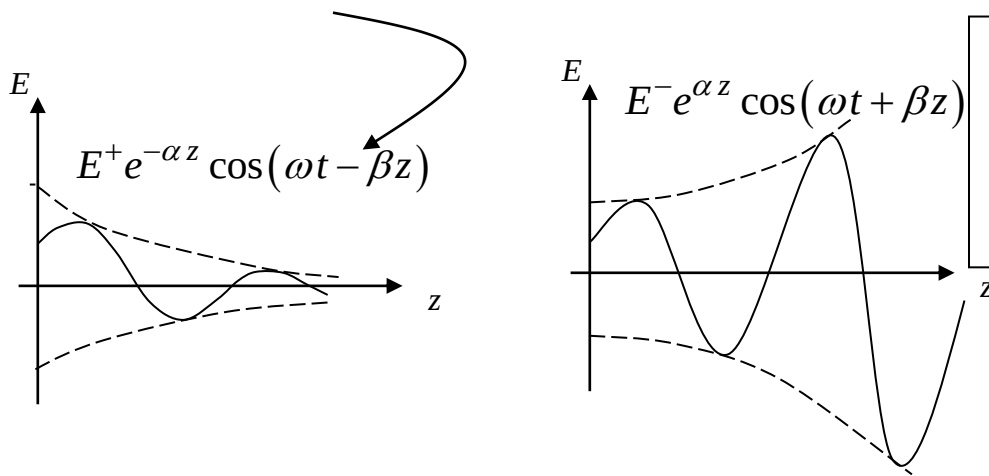
$$\nabla^2 \mathbf{E} + k_s^2 \mathbf{E} = 0$$

Typical Solution in frequency domain with loss $k_s = -j\gamma$ where $\gamma = \alpha + j\beta$

$$\begin{aligned} E_x &= E^+ e^{-\gamma z} + E^- e^{\gamma z} \\ &= E^+ e^{-\alpha z} e^{-j\beta z} + E^- e^{\alpha z} e^{j\beta z} \end{aligned}$$

In the time domain this becomes

$$\begin{aligned} \mathcal{E}_x(t) &= \text{Re}\{E_x e^{j\omega t}\} \\ &= \underbrace{E^+ e^{-\alpha z} \cos(\omega t - \beta z)} + E^- e^{\alpha z} \cos(\omega t + \beta z) \end{aligned}$$



Poynting Theorem:

$$\underbrace{\nabla \cdot (\mathbf{E} \times \mathbf{H}^*)}_{\substack{\doteq \mathbf{S} \\ \text{vol. power} \\ \text{leaving a point}}} + \underbrace{\sigma E^2}_{\substack{\doteq Pd \\ \text{dissipated} \\ \text{vol. power} \\ \text{density}}} + j2\omega \underbrace{\left(-\frac{1}{2} \epsilon E^2 + \frac{1}{2} \mu H^2 \right)}_{\substack{-w_e \quad w_m \\ \text{stored electric} \\ \text{and magnetic} \\ \text{energy} \\ \text{density}}} = - \underbrace{\mathbf{E} \cdot \mathbf{J}^{i*} - \mathbf{H}^* \cdot \mathbf{M}^i}_{\substack{\text{power} \\ \text{density} \\ \text{supplied} \\ \text{by the} \\ \text{sources}}}$$

Plane waves

Wave polarization

$$\begin{aligned} \mathbf{E} &= (E_{x0} \hat{\mathbf{x}} + E_{y0} \hat{\mathbf{y}}) e^{-jkz} \\ &= (|E_{x0}| e^{j\theta_x} \hat{\mathbf{x}} + |E_{y0}| e^{j\theta_y} \hat{\mathbf{y}}) e^{-jkz} \end{aligned}$$

In the time domain

$$\mathbf{E}(t) = |E_{x0}| \cos(\omega t - kz + \theta_x) \hat{\mathbf{x}} + |E_{y0}| \cos(\omega t - kz + \theta_y) \hat{\mathbf{y}}$$

Case 1 Linear Polarization

Circular Polarization

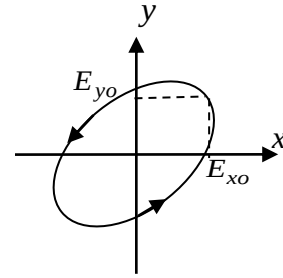
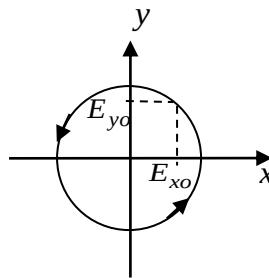
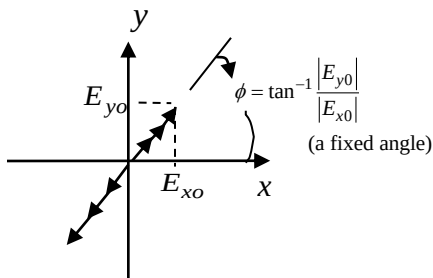
Elliptical Polarization

:

Let $E_{x0} \neq E_{y0}$, $\theta_x = \theta_y$

$|E_{x0}| = |E_{y0}| = E_0$, $\theta_x = \theta_y \pm \pi/2$

$|E_{x0}| \neq |E_{y0}|$, $\theta_x \neq \theta_y$ and $\theta_x \neq \theta_y \pm \pi/2$



Boundary Conditions

scalar form

$$E_{t1} - E_{t2} = M_s$$

$$H_{t1} - H_{t2} = J_s$$

$$D_{n1} - D_{n2} = \rho_s$$

$$B_{n1} - B_{n2} = \rho_{ms}$$

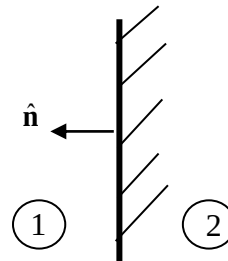
vector form

$$\hat{\mathbf{n}} \times (\mathbf{E}_1 - \mathbf{E}_2) = \mathbf{M}_s$$

$$\hat{\mathbf{n}} \times (\mathbf{H}_1 - \mathbf{H}_2) = \mathbf{J}_s$$

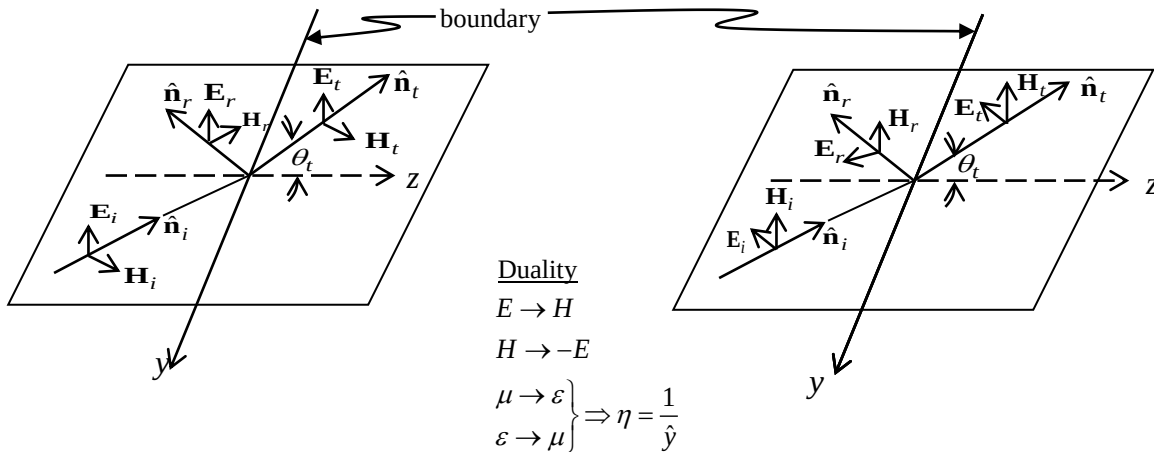
$$\hat{\mathbf{n}} \cdot (\mathbf{D}_1 - \mathbf{D}_2) = \rho_s$$

$$\hat{\mathbf{n}} \cdot (\mathbf{B}_1 - \mathbf{B}_2) = \rho_{ms}$$



⊥ polarization

|| polarization



Used to derive:

Snell's law - $k_{y1} = k_{y2}$ propagation constants parallel to a boundary must always be equal

Reflection and Transmission Coefficients - Γ and T

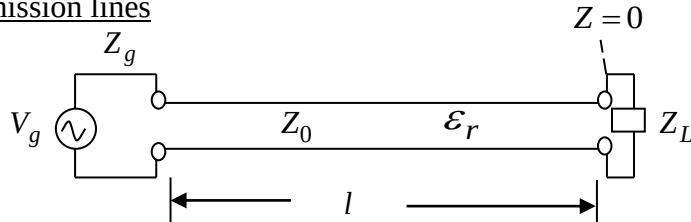
Critical Angle – angle of total reflection $T = 0$ or set $\beta_2 = 0$

Brewster Angle - angle of total transmission $\Gamma = 0$

Evanescent (or surface) wave - wave trapped on a dielectric interface when $\theta_i > \theta_c$

Depth of penetration – distance at which evanescent wave drops to $1/e$ of initial value.

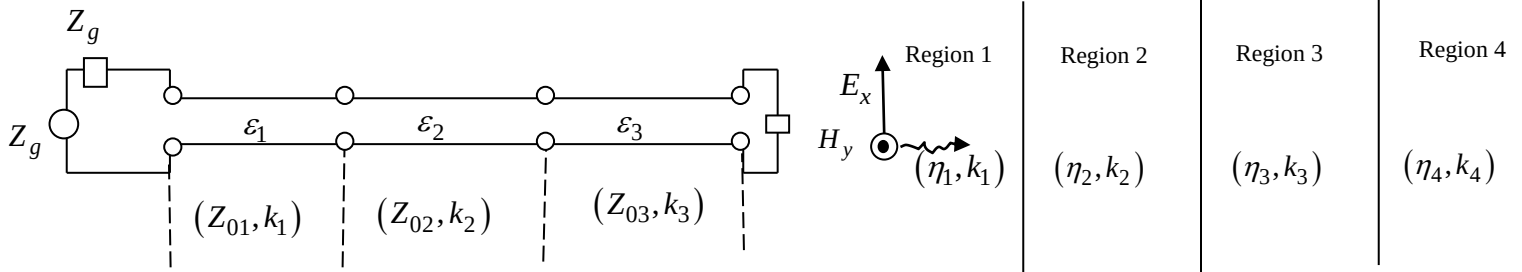
Transmission lines



Impedance matching

Open Circuit, Short Ckt etc

Transmission line duals



Potentials

$$\nabla^2 \mathbf{A} + k^2 \mathbf{A} = -\mu \mathbf{J}$$

$$\mathbf{E} = -\frac{j\omega}{k^2} (k^2 \mathbf{A} + \nabla \nabla \cdot \mathbf{A})$$

$$\mathbf{H} = \frac{1}{\mu} \nabla \times \mathbf{A}$$

The potential for free space with an elemental source

$$\mathbf{A}(\mathbf{r}) = \frac{\mu I l}{4\pi} \frac{e^{-jk|\mathbf{r}-\mathbf{r}'|}}{|\mathbf{r}-\mathbf{r}'|} \hat{\mathbf{z}}$$

The potential for free space with a general current distributions

$$\mathbf{A} = \frac{\mu}{4\pi} \int_{V'} \mathbf{J}(\mathbf{r}') \frac{e^{-jk|\mathbf{r}-\mathbf{r}'|}}{|\mathbf{r}-\mathbf{r}'|} dV'$$

Which cannot be solved analytically so we use

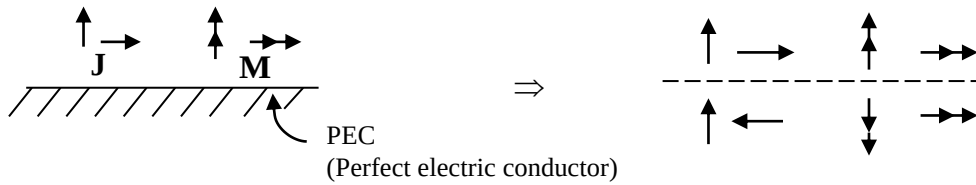
$$|\mathbf{r}-\mathbf{r}'| \rightarrow r - r' \cos \xi$$

$$r' \cos \xi = (x' \cos \phi + y' \sin \phi) \sin \theta + z' \cos \theta$$

$$r' \cos \xi = \rho' \sin \theta \cos(\phi - \phi') + z' \cos \theta$$

$$r' \cos \xi = r' [\cos \theta \cos \theta' + \sin \theta \sin \theta' \cos(\phi - \phi')]$$

Method of Images



Theorems of Electromagnetics

Image methods

Reciprocity

Equivalence principal

Transverse Resonance method

Construction of solutions to waveguide and scattering problems

TM Solutions

$$\nabla^2 \mathbf{A} + k^2 \mathbf{A} = 0 \quad (5a)$$

$$\mathbf{E} = \frac{-j\omega}{k^2} (k^2 \mathbf{A} + \nabla \nabla \cdot \mathbf{A}) \quad (6a)$$

$$\mathbf{H} = \frac{1}{\mu} \nabla \times \mathbf{A} \quad (7a)$$

$$\mathbf{A} = \psi^{TM} \hat{\mathbf{u}} \quad (8a)$$

TM case

$$\mathbf{A} = \psi^{TM} \hat{\mathbf{u}}$$

$$\mathbf{H} = \frac{1}{\mu} \nabla \psi^{TM} \times \hat{\mathbf{u}}$$

$$\mathbf{E} = -\frac{j\omega}{k^2} \left(k^2 \psi^{TM} \hat{\mathbf{u}} + \nabla \frac{\partial}{\partial u} \psi^{TM} \right)$$

$$\nabla^2 \psi^{TM} + k^2 \psi^{TM} = 0$$

TE case

Similarly let $\mathbf{F} = \psi^{TE} \hat{\mathbf{u}} \Rightarrow$

$$\mathbf{E} = \frac{1}{\varepsilon} \hat{\mathbf{u}} \times \nabla \psi^{TE} \quad (9)$$

$$\mathbf{H} = \frac{j\omega}{k^2} \left(k^2 \psi^{TE} \hat{\mathbf{u}} + \nabla \frac{\partial}{\partial u} \psi^{TE} \right) \quad (10)$$

$$\nabla^2 \psi^{TE} + k^2 \psi^{TE} = 0$$

TE Solutions

$$\nabla^2 \mathbf{F} + k^2 \mathbf{F} = 0 \quad (5b)$$

$$\mathbf{E} = -\frac{1}{\varepsilon} \nabla \times \mathbf{F} \quad (6b)$$

$$\mathbf{H} = \frac{-j\omega}{k^2} (k^2 \mathbf{F} + \nabla \nabla \cdot \mathbf{F}) \quad (7b)$$

$$\mathbf{F} = \psi^{TE} \hat{\mathbf{u}} \quad (8b)$$

Algorithm for solving boundary value problems

